PHYSICS

Paper 1 (Theory)

Maximum Marks: 70 **Time Allowed: Three Hours**

Reading Time: Additional Fifteen Minutes

Instructions to Candidates

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during reading time.
3. This question paper has 9 printed pages.
4. It has twenty questions in this paper. Answer all questions.
5. There are four sections in the paper: A, B, C and D. Internal choices have been provided in two questions each in Sections B, C and D.
6. Section A consists of one question having fourteen subparts of one mark each.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B consists of seven questions of two marks each.
9. Section C consists of nine questions of three marks each.
10. Section D consists of three questions of five marks each.
11. The intended marks for questions are given in brackets [ ].
12. A list of useful constants and relations is given below these instructions.
13. A simple scientific calculator without a programmable memory may be used for calculations.

USEFUL CONSTANTS AND RELATIONS

1.	Planck's constant	h	=	$6{\cdot}6 \times 10^{-34}$ J s
2.	Speed of light in vacuum	c	=	3×10^{8} m s^{-1}
3.	Charge of a proton	e	=	$1{\cdot}6 \times 10^{-19}$ C
4.	Mass of an electron	m_e	=	$9{\cdot}1 \times 10^{-31}$ kg
5.	Permittivity of vacuum	ε_0	=	$8{\cdot}85 \times 10^{-12}$ F m^{-1}
6.	Permeability of vacuum	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
7.	One atomic mass unit	1 u	=	$1{\cdot}66 \times 10^{-27}$ kg = 931·5 MeV/c^2
8.	One electron volt	1 eV	=	$1{\cdot}6 \times 10^{-19}$ J
9.		1 MeV	=	$1{\cdot}6 \times 10^{-13}$ J
10.	Rydberg constant	R	=	$1{\cdot}097 \times 10^{7}$ m^{-1}

Note: The questions in this sample paper follow the format of the ISC Specimen Question Paper 2027. The weightage allocated to each unit in the syllabus has been strictly adhered to.

SECTION A – 14 MARKS

Question 1

(A) In questions (i) to (vii), choose the correct alternative (a), (b), (c) or (d) for each of the questions given below.

(i) Point charges of +30 nC, −10 nC and +5 nC lie inside a large closed spherical surface, and a further charge of +40 nC lies outside it. The net electric flux through the surface is: *(Apply)* [1]

(a) $5{\cdot}1 \times 10^3$ N m^2 C^{-1}

(b) $2{\cdot}8 \times 10^3$ N m^2 C^{-1}

(c) $7{\cdot}3 \times 10^3$ N m^2 C^{-1}

(d) zero

(ii) When the current in a coil changes from 9 A to 3 A in 4×10^{-2} s, the emf induced in the coil is 3 V. The self-inductance of the coil, in millihenry, is: *(Apply)* [1]

(a) 5

(b) 10

(c) 20

(d) 40

(iii) In a series LCR circuit, match the condition in Column I with the behaviour in Column II. *(Recall)* [1]

Column I	Column II
1. $\omega L > 1/\omega C$	A. The current is in phase with the voltage
2. $\omega L < 1/\omega C$	B. The current lags behind the voltage
3. $\omega L = 1/\omega C$	C. The current leads the voltage

(a) 1-B, 2-C, 3-A

(b) 1-C, 2-B, 3-A

(c) 1-B, 2-A, 3-C

(d) 1-A, 2-B, 3-C

(iv) An α-particle (mass m, charge $2e$) is accelerated from rest through a potential difference V. Its de Broglie wavelength is: *(Apply)* [1]

(a) $h/\sqrt{(2meV)}$

(b) $h/\sqrt{(4meV)}$

(c) $h/\sqrt{(8meV)}$

(d) $2h/\sqrt{(2meV)}$

(v) Select the odd one out from the following dopants. *(Analyse)* [1]

(a) Boron

(b) Indium

(c) Gallium

(d) Arsenic

(vi) Given below are two statements marked, Assertion and Reason. Read the two statements and choose the correct option. [1]

Assertion: The graph of v against u for the real images formed by a concave mirror lies in the third quadrant.

Reason: For a real image formed by a concave mirror, both u and v are negative in the Cartesian sign convention. *(Analyse)*

(a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Assertion is false and Reason is true.

(vii) Monochromatic light is used in Young's double slit experiment and then in a single-slit diffraction experiment. Identify the correct statements. *(Understand)* [1]

(P) The bright fringes of the double-slit pattern are of equal width.

(Q) In the single-slit pattern the central maximum is the widest and the brightest.

(R) All the maxima of the single-slit pattern are of equal intensity.

(S) The central fringe of the double-slit pattern is twice as wide as the others.

(a) Both (Q) and (R)

(b) Both (P) and (S)

(c) Both (R) and (S)

(d) Both (P) and (Q)

(B) Answer the following questions briefly:

(i) A charge of 3 nC is moved through 5 cm along the perpendicular bisector (equatorial line) of an electric dipole made of charges $+Q$ and $-Q$ placed 2 cm apart. Calculate the work done. *(Apply)* [1]

(ii) A 24 V battery, a 6 Ω resistor and an ammeter are connected in series. The ammeter has a coil of resistance 180 Ω shunted by 20 Ω. Find the reading of the ammeter. *(Apply)* [1]

(iii) State one advantage of a reflecting telescope over a refracting telescope. *(Recall)* [1]

(iv) Fill in the blank with an appropriate term from the bracket. [1]

If the separation of the two slits in Young's double slit experiment is doubled, the fringe width _____ (is halved, is doubled, remains the same). *(Understand)*

(v) In an experiment on the photoelectric effect, the slope of the graph of cut-off voltage against frequency of the incident light is found to be $4{\cdot}1 \times 10^{-15}$ V s. Calculate the value of Planck's constant. *(Apply)* [1]

(vi) Name the particles and the radiation involved in pair production, and write the equation relating them. *(Recall)* [1]

(vii) Name the diode that is used as a voltage regulator and state the bias in which it is operated. *(Recall)* [1]

SECTION B – 14 MARKS

Question 2

(i) The electric potential V varies with distance x as shown in Figure 1. Construct a graph of the electric field strength E as a function of x. *(Analyse)* [2]

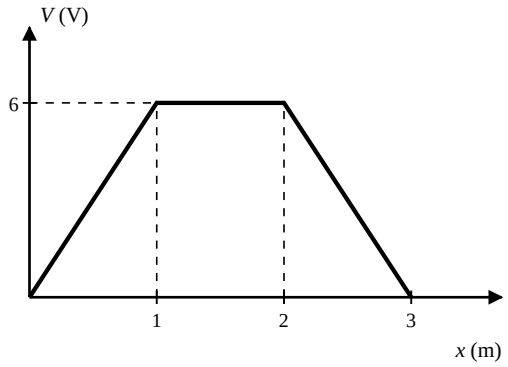


Figure 1

OR

(ii) Figure 2 shows a closed surface S and four point charges. Calculate the electric flux emanating from the closed surface S. *(Apply)* [2]

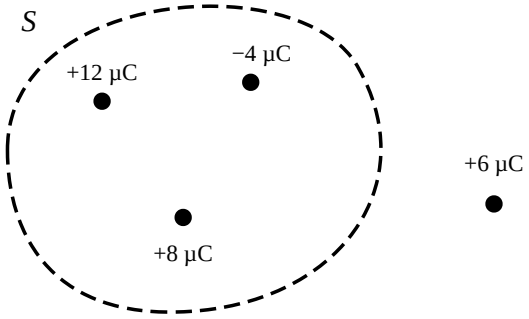


Figure 2

Question 3

Three identical capacitors, each of capacitance 6 µF, are connected in parallel across a 12 V battery. (a) Calculate the energy stored in the system. (b) Find the ratio C_s/C_p of the equivalent capacitance when the three are connected in series to that when they are connected in parallel. *(Apply)* [2]

Question 4

(i) A heater takes 20 minutes to boil a given amount of water. Its coil is cut and one-fourth of its length is used in the heater. How much time will it now take to boil the same amount of water using the same source, starting from the same initial temperature? *(Apply)* [2]

OR

(ii) (a) Name a material that has a very low temperature coefficient of resistance. (b) Give an example of a non-Ohmic conductor and state how its *V–I* graph differs from that of an Ohmic one. *(Recall)* [2]

Question 5

(a) State the SI unit of the magnetic dipole moment of a current-carrying loop and the direction of the moment. (b) Calculate the magnetic moment of a circular coil of 100 turns and radius 5 cm carrying a current of 2 A. *(Apply)* [2]

Question 6

(i) A convex lens of focal length 25 cm is in contact with a concave lens of focal length 40 cm. Calculate the power and the focal length of the combination. *(Apply)* [1]

(ii) The angle of a glass prism is 60° and the angle of minimum deviation is 38°. Without using the refractive index of the glass, find the angle of incidence of the ray suffering minimum deviation. *(Apply)* [1]

Question 7

Analyse the following two situations and name the electromagnetic wave responsible in each case. *(Analyse)*

(i) The glass of a greenhouse lets the sunlight in but does not let the heat radiated by the warm soil and plants escape. Which radiation is trapped? *(Analyse)* [1]

(ii) A welder wears a protective mask or goggles while the arc is struck. Name the radiation from the arc against which the eyes need protection. *(Analyse)* [1]

Question 8

The work functions of two metals M_1 and M_2 are 2·2 eV and 4·5 eV respectively. Monochromatic light of wavelength 500 nm falls on each. Perform the necessary calculations to find which metal emits photoelectrons, and the maximum kinetic energy of the ejected electrons. *(Apply)* [2]

SECTION C – 27 MARKS

Question 9

(i) Show that the electric field at a point on the axial line of a short electric dipole, at a distance r from its centre, is given by $E = \frac{1}{4\pi\varepsilon_0}\frac{2pr}{(r^2 - l^2)^2}$, where the symbols have their usual meaning. *(Understand)* [3]

OR

(ii) An infinite plane sheet carries a uniform surface charge density +σ. Using Gauss's law, find the electric field at a point at a distance r from it. Hence write the resultant field at a point **between** two such parallel sheets carrying +σ and −σ, and give its direction. *(Understand)* [3]

Question 10

(i) In the network of Figure 3 the cells have negligible internal resistance and the current through the 4 Ω resistor is 1·5 A. Using Kirchhoff's laws, find (a) the current I_1, (b) the current I_2 and the emf E of the second cell. *(Apply)* [3]

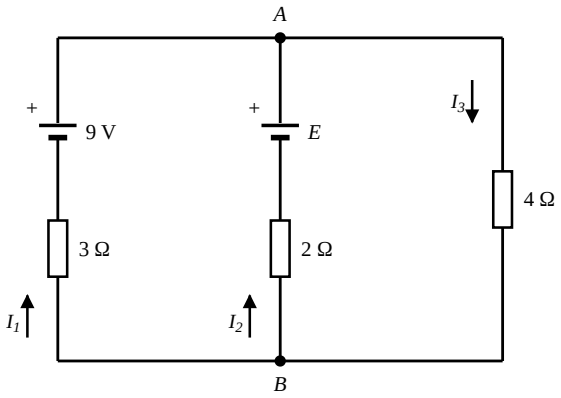


Figure 3

OR

(ii) In the metre bridge of Figure 4, the resistance in the left gap is 2 Ω and the null point is found at 40 cm from the left end. What resistance should be connected to the unknown resistance X, and how, so that the null point shifts to 50 cm from the left end? *(Analyse)* [3]

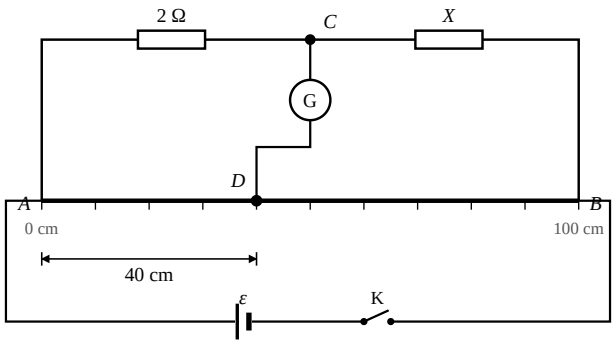


Figure 4

Question 11

State Ampère's circuital law. Using it, obtain an expression for the magnetic flux density B at a point X at a perpendicular distance r from a long straight conductor carrying a current I. *(Understand)* [3]

Question 12

(i) Two long straight parallel wires 10 cm apart carry currents of 5 A and 8 A in the same direction. Calculate the force per unit length between them and state whether it is attractive or repulsive. *(Apply)* [1]

(ii) A galvanometer of resistance 50 Ω gives full-scale deflection for a current of 2 mA. How can it be converted into a voltmeter of range 0–20 V? Find the resistance of the voltmeter. *(Apply)* [2]

Question 13

An optics student uses a compound microscope. The examiner gives instructions and asks questions. State whether the response of the student in each case is correct or incorrect. Give a reason for your answer. *(Evaluate)*

(i) **Examiner:** Increase the magnification produced by the objective lens. [1]
Student: Decrease the distance between the object and the objective, keeping in mind that this distance must remain greater than the focal length of the objective.

(ii) **Examiner:** Increase the resolving power of the instrument. [1]
Student: Illuminate the object with light of a longer wavelength.

(iii) **Examiner:** Which of the lenses of focal lengths $f_1 = 1{\cdot}5$ cm, $f_2 = 80$ cm and $f_3 = 5$ cm would you take as the objective and the eyepiece? [1]
Student: $f_1 = 1{\cdot}5$ cm as the objective and $f_2 = 80$ cm as the eyepiece.

Question 14

Using Huygens' wave theory and a labelled diagram, show that the angle of reflection is equal to the angle of incidence. *(Understand)* [3]

Question 15

For a prism of refracting angle A, show that the refractive index of its material is $n = \sin[(A + \delta_m)/2]/\sin(A/2)$, where δ_m is the angle of minimum deviation. *(Understand)* [3]

Question 16

(i) Draw a labelled ray diagram of an astronomical telescope when its final image lies at infinity. *(Understand)* [1]

(ii) An astronomical telescope has an objective of focal length 150 cm and an eyepiece of focal length 5 cm. Calculate (a) its magnifying power and the length of the tube in normal adjustment, and (b) the magnifying power when the final image is formed at the least distance of distinct vision (25 cm). *(Apply)* [2]

Question 17

(i) A monochromatic source emits light of wavelength 248 nm. Calculate the energy and the momentum of each photon. *(Apply)* [2]

(ii) Plot a labelled graph of the stopping potential V_s against the frequency ν of the incident radiation, and state how Planck's constant is obtained from it. *(Understand)* [1]

SECTION D – 15 MARKS

Question 18

(i)(a) A resistor *R*, an inductor *L* and a capacitor *C* are connected in series to an a.c. source. Using a phasor diagram, derive an expression for the impedance of the circuit. *(Understand)* [3]

(b) A series LCR circuit has $R = 10\ \Omega$, $L = 5$ mH and $C = 20\ \mu$F and is connected to a 220 V a.c. source of variable frequency. (1) Draw a graph of the current *I* against the angular frequency ω and label the resonant frequency, the current amplitude and the bandwidth. (2) Find the resonant frequency and the current at resonance. *(Apply)* [2]

OR

(ii)(a) A coil is connected to a 100 V d.c. supply and carries a current of 2 A. When it is connected to a 100 V, 50 Hz a.c. supply the current is 1 A. (1) Explain why the current is less with the a.c. supply. (2) Calculate the coefficient of self-inductance of the coil. *(Apply)* [3]

(b) A rod of length 0·4 m slides at a constant speed of 3 m s^{-1} over two parallel rails joined by a 5 Ω resistor, in a uniform magnetic field of 0·25 T perpendicular to the plane of the rails. Calculate (1) the induced emf and the current, and (2) the force needed to keep the rod moving at this speed. *(Apply)* [2]

Question 19

(i)(a) Using Bohr's postulates, derive an expression for the orbital velocity of the electron in the *n*th orbit of a hydrogen-like atom of atomic number *Z*. *(Understand)* [3]

(b) Hence show that the speed of the electron in the first orbit of hydrogen is about 1/137 of the speed of light. *(Apply)* [1]

(c) A hydrogen atom goes from the third excited state to the second excited state. Calculate the wavelength of the radiation emitted and name the series and the region of the spectrum to which it belongs. *(Apply)* [1]

OR

(ii)(a) Sketch a graph showing the variation of the binding energy per nucleon of a nucleus with its mass number. Mark on it the regions of nuclei that are (1) stable, (2) prone to fission, (3) prone to fusion. *(Understand)* [3]

(b) Calculate the energy released in the nuclear reaction $^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He}$ + energy. (Mass of $^{2}_{1}\mathrm{H} = 2{\cdot}014102$ u, mass of $^{4}_{2}\mathrm{He} = 4{\cdot}002604$ u, 1 u = 931·5 MeV/c^2) *(Apply)* [2]

Question 20

A rectifier is an electronic circuit that converts alternating current into direct current. Ravi builds one for his school laboratory. He starts with a p–n junction diode and studies how the depletion region changes with the bias. A half-wave rectifier uses a single diode, so only one half of every cycle reaches the load and its output is a poor, pulsating d.c. Ravi therefore builds a full-wave rectifier from two diodes and a centre-tapped transformer, powered by the 50 Hz mains, and compares the two outputs. Finally he uses a Zener diode to keep the output voltage constant.

(i) State how the thickness of the depletion region of a p–n junction changes in forward bias and in reverse bias. *(Recall)* [1]

(ii) Draw a neat, labelled circuit diagram of the full-wave rectifier built by Ravi. *(Understand)* [1]

(iii) Show the input and the output waveforms for this rectifier. *(Understand)* [1]

(iv) What are the frequencies of the ripple in the output of the half-wave and of the full-wave rectifier for a 50 Hz input? *(Apply)* [1]

(v) A Zener diode of breakdown voltage 6·0 V is connected in the reverse direction across the output through a 300 Ω series resistor from a 12 V supply (no load connected). Find the current through the series resistor. *(Apply)* [1]

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